



Week 1 Lecture 9

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Performance of a Website

- How fast a site can be?
- What limits the performance?
- Some basic observations ...

Latency

How much time does it take to get the response for a given request

- Speed of light: $3 \times 10^8 m/s$ in vacuum
- $\sim 2 \times 10^8 m/s$ on a cable
 - $\sim 5ns$ for 1m $\implies \sim 5ms$ for 1000km
- Datacenter is 2000km away
 - One-way request: $\sim 10ms$
 - Round-trip: 20ms

So, what does this mean?

If I am continuously sending requests to a server, and I want it such that whenever I send a request, I get a response back before I can then proceed, then

I am limited to a maximum of 50 requests per second

What about a server that needs to send back responses?

Response size

- Response: 1KB of text (headers, HTML, CSS, JS)
- Network connection = 100Mbps (Megabits per second) $\approx$ 10MBps (Megabytes per second) [Theoretically, it should be 12.5MB/s]
- This limits us to $\sim$ 10,000 requests per second

For eg: Just imagine, that the result of some entrance exams were declared in India and if it is all trying to come out of one server with the bandwidth of 100 Megabits per second

The server would simply crash when it is bombarded with more than 10,000 requests per second

A simple case study: Google response

- Headers: $\sim$ 100B (bytes)
- Content: 144kB (kilobytes)
- Approx. 60,000 requests per second

Which means that Google server effectively need more than 80Gbps of bandwidth

Memory: YouTube

- One python HTTP server process: $\sim$ 6MB (measured)
- Multiple parallel requests: multiple processes
 - eg: YouTube will have long running server processes
- 2016 Presidential debate in the US:
 - 2M+ concurrent viewers

Storage: Google

- Index 100s of billions of web pages
- Cross-reference, [pagerank](#)
- Total index size estimate:
 - 100,000,000 Gigabytes $\implies$ 100 Petabytes
- Storage?
- Retrieval

Summary

- The Web is a useful device / OS agnostic platform for apps
- Built for HTTP for transport, HTML and related tech for presentation
- Servers trivial at most basic
- Scaling requires careful design